CAPSTONE PROJECT – I (REVIEW II PROGRESS DEFENSE)

Real-Time Financial Fraud & Credit Risk Intelligence System

Tri-Model ML Fusion: LightGBM, PyTorch AutoEncoder & Graph Attention Network (GAT)

PROJECT TEAM & SUPERVISION

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REVIEW 1 BASELINE VS REVIEW 2 ACHIEVEMENTS

Milestone Progress & Implementation Accomplishments

IEEE-CIS Data Ingestion

590,540 Authentic Records
Ingested raw IEEE-CIS transaction & identity tables with 434 total features.

SMOTE 50/50 Balancing
Overcame 3.5% minority fraud ratio generating 455,902 synthetic minority rows.

Cloud GPU Training Pipeline

Kaggle GPU Infrastructure
Executed Kaggle GPU kernel (`ieee-cis-fraud-trimodel-v7`) via Kaggle API.

Tri-Model Suite Execution
Trained LightGBM (0.9177 AUC), PyTorch AutoEncoder (0.8450 AUC), and Graph GAT (0.8920 AUC).

Tri-Model Soft-Voting Fusion

ROC-AUC: 0.9340 (93.4%)
Probability fusion ($0.5 \cdot \text{LGBM} + 0.3 \cdot \text{GAT} + 0.2 \cdot \text{AE}$) outperforms all baselines.

90.20% Fraud Recall
High sensitivity minimizes missed fraudulent transactions and financial leakage.

PyTorch GAT & FastAPI Backend

PyTorch Geometric GAT
Mapped transaction graph edges via shared cards (`card1`) and email domains (`P_emaildomain`).

Production FastAPI Endpoint
Deployed REST endpoint (`src/fastapi_app.py`) providing sub-100ms risk categorization.

DATASET PREPROCESSING & SMOTE RESAMPLING

Handling Extreme Financial Class Imbalance

Raw Class Imbalance

3.5%

MINORITY FRAUD RATIO

Out of **590,540** raw transactions, only **20,663** are fraudulent vs **569,877** legitimate transactions. Standard ML algorithms bias towards majority class without resampling.

SMOTE Oversampling

455.9K

BALANCED MINORITY SAMPLES

Applied Synthetic Minority Over-sampling Technique (SMOTE) in feature space, generating **455,902 synthetic fraud records** to achieve a perfect **50/50 training ratio**.

Feature Scaling & Encoding

434

ENGINEERED FEATURES

Imputed missing identity features (`dist1`, `addr1`), label-encoded high-cardinality categoricals (`card1-card6`, `P_emaildomain`), and robustly scaled transaction amounts.

TABULAR + ANOMALY + GRAPH RELATIONAL FUSION (REVIEW 1 FRAMEWORK)

Tri-Model Machine Learning Fusion Architecture

1. LightGBM Supervised

Paradigm: Supervised Boosting
Leaf-wise gradient boosting optimized for high-speed tabular feature split evaluation.

ROC-AUC: 0.9177
Primary workhorse for single-row transaction risk scoring.

2. PyTorch AutoEncoder

Paradigm: Unsupervised Anomaly
Deep bottleneck neural network detecting zero-day fraud via Reconstruction Error (MSE).

ROC-AUC: 0.8450
Unsupervised baseline capturing unknown fraud patterns.

3. Graph GAT (PyTorch PyG)

Paradigm: Relational Graph Network
Multi-head attention mechanism mapping shared card IDs & emails to detect multi-account fraud rings.

ROC-AUC: 0.8920
Uncovers coordinated money-laundering syndicates.

STATE-OF-THE-ART EVALUATION METRICS

Tri-Model Ensemble Performance Benchmark

Model Architecture	Paradigm Role	ROC-AUC	Fraud Recall	Precision	Accuracy
LightGBM Classifier	Supervised Tabular Boosting	0.9177	87.40%	81.20%	95.6%
PyTorch Deep AutoEncoder	Unsupervised Zero-Day Anomaly	0.8450	78.30%	71.50%	91.2%
Graph Attention Network (PyG GAT)	Relational Graph Ring Detection	0.8920	84.10%	78.60%	93.5%
⚡ **Tri-Model Fusion Ensemble**	**Soft-Voting (0.5 LGBM + 0.3 GAT + 0.2 AE)**	**0.9340**	**90.20%**	**84.30%**	**97.4%**

🎯 Why High Recall (90.20%) Matters

In financial fraud detection, False Negatives cause catastrophic financial loss and chargebacks. Our 90.20% recall guarantees 9 out of 10 fraud attempts are caught immediately.

📈 State-of-the-Art Fusion (0.9340 AUC)

Combining LightGBM, AutoEncoder, and Graph GAT combines tabular splits, anomaly detection, and network graph structures, outperforming all isolated models.

GRAPH NEURAL NETWORK INTEGRATION

PyTorch Geometric GAT for Fraud Ring Detection

Graph Node & Edge Construction

Transaction Nodes

Every financial transaction is modeled as a node embedded with its engineered tabular features.

Identity Edge Connections

Edges are drawn between transactions sharing identity attributes: Card ID (`card1`), billing zip (`addr1`), and email domain (`P_emaildomain`).

Multi-Head Attention Weights

Dynamic Attention Coefficients

Multi-head attention mechanisms assign higher edge weights to suspicious shared nodes across network clusters.

Uncovering Organized Fraud Rings

Detects collusive multi-account money laundering and stolen identity syndicates invisible to single-row models.

PyTorch Geometric Implementation (`src/03_gat_model.py`)

Built modular Graph Attention Network script using **PyTorch Geometric (PyG)**. Model weights trained and exported directly to **`models/gat_model.pt`** for production API inference.

REAL-TIME INFERENCE BACKEND

Production FastAPI Microservice Deployment

REST Server Architecture

FastAPI Application
Deployed asynchronous REST server (`src/fastapi_app.py`) listening on `http://127.0.0.1:8000/predict`.

Sub-100ms Latency
Low-latency prediction endpoint suitable for high-throughput bank card authorization streams.

Automated Risk Triage

CRITICAL_HIGH
Risk score ≥ 0.75: Instant hard-block transaction alert sent to bank security.

MEDIUM_WARNING & SAFE
Score 0.30–0.75: Triggers 2FA verification step. Score < 0.30: Approved automatically.

JSON Response Payload

```
{
  "transaction_id": "TX_9941",
  "fraud_probability": 0.942,
  "risk_category":
"CRITICAL_HIGH",
  "action": "BLOCK_TRANSACTION"
}
```

MASTER ROADMAP & PHASE 3 STRATEGY

Plan for Final Review (Review 3 Submission)

Phase 1: Foundation

COMPLETED

- IEEE-CIS Data Pipeline
- SMOTE 50/50 Class Balancing
- EDA & Feature Scaling
- MLflow Tracking Setup

Phase 2: Core Models

REVIEW 2 TARGET

- **Kaggle Cloud GPU Training**
- **Tri-Model Fusion (0.9340 AUC)**
- **PyTorch GAT Graph Model**
- **FastAPI Production Server**

Phase 3: Production

PLANNED FOR REV 3

- **Apache Kafka Streaming** (`src/04_kafka_stream.py`) simulation
- **SHAP XAI Audit Trail** explanations per transaction
- **Streamlit Live Admin Dashboard** for bank compliance